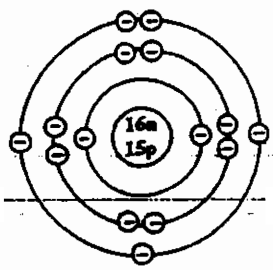
CSEC Chemistry
June 2006 Paper 1

1. Sulphur and oxygen are in the same group of the periodic table because

(A) they can react with each other
(B) the atomic number of sulphur is 16 and the relative atomic mass of oxygen 16
(C) they have the same number of electrons in their outer shell
(D) they can form covalent compounds

Item 2 refers to the following diagram which represents the structure of atom X.



2. If atom X forms an ion, the charge on the ion would MOST likely be

(A) 5^+
(B) 5^-
(C) 3^+
(D) 3^-

3. Which of the following oxides does NOT react with acids to form only a salt and water?

(A) PbO
(B) CaO
(C) ZnO
(D) MnO_2

4. Which two of the following statements about covalent compounds are true?

I. They are mainly solids.
II. They usually have low melting and boiling points.
III. They are usually insoluble in water.
IV. They contain only ions.

(A) I and III
(B) I and IV
(C) II and III
(D) II and IV

5. Which of the following statements illustrates osmosis?

(A) Pollen dust in random motion in water
(B) Movement of perfume molecules through the air in a room
(C) Movement of water molecules across a cellophane membrane into a concentrated glucose solution
(D) Movement of food colouring molecules in a dilute aqueous solution

6. Radioactive isotopes are NOT normally used in the

(A) determination of the age of fossils
(B) treatment of cancer
(C) treatment of influenza
(D) powering of certain types of submarines

7. PO_4^{3-} is the formula for the phosphate anion. What is the formula of calcium phosphate? (Calcium is element number 20.)

(A) $Ca_3(PO_4)_2$

(B) $Ca_2(PO_4)_3$

(C) Ca_2PO_4

(D) $Ca(PO_4)_2$

8 From which of the following can a solid be obtained by the process of sedimentation?

(A) Gels
(B) Emulsions
(C) Foams
(D) Suspensions

9. In the reaction between zinc and dilute sulphuric acid

(A) water is formed
(B) an insoluble salt is formed
(C) neutralisation occurs
(D) oxidation and reduction occur

10. Which of the following elements exhibits allotrophy?

(A) Carbon
(B) Uranium
(C) Butane
(D) Silicon

11. Which of the following may be defined as that quantity of a substance which contains 6.02 x 10^{23} particles of that substance?

(A) Atom
(B) Mole
(C) Formula
(D) Molecule

12. The values of x and y respectively in the equation

$x\,KOH + 3\,Br_2 \rightarrow y\,KBr + KBrO_3 + 3\,H_2O$

are

(A) $x = 5; y = 6$
(B) $x = 6; y = 5$
(C) $x = 5; y = 5$
(D) $x = 6; y = 6$

13. How many covalent bonds are there in a nitrogen molecule?

(A) 4
(B) 3
(C) 2
(D) 1

14. A separating funnel can be used to separate a mixture of

(A) water and sodium chloride
(B) water and ethanol
(C) water and kerosene
(D) kerosene and solid sodium chloride

15. A substance X, with a boiling point of 58 °C, is miscible with another liquid Y, of boiling point 94 °C. A mixture of these two liquids can BEST be separated into its components by

(A) fractional distillation
(B) simple distillation
(C) use of a separating funnel
(D) sublimation

16. Which of the following gases is composed of molecules containing two atoms each?

(A) Hydrogen chloride
(B) Carbon dioxide
(C) Ammonia
(D) Methane

<u>Item 17</u> refers to the following information.

A compound when treated with concentrated hydrochloric acid gives off a green gas which bleaches moist red litmus paper.

17. The gas given off is

(A) hydrogen sulphide
(B) chlorine
(C) sulphur dioxide
(D) nitrogen dioxide

18. In which of the following substances will electrolysis NOT occur when an electrical current is passed through it?

(A) Sodium hydroxide solution
(B) Molten sodium chloride
(C) Solid sodium chloride
(D) Dilute sulphuric acid

19. Sulphuric acid is a stronger acid than ethanoic acid (acetic acid) in aqueous solution because sulphuric acid

(A) is obtainable in high concentrations
(B) ionises to a greater extent than ethanoic acid
(C) is more corrosive than ethanoic acid
(D) causes sugar to char whereas ethanoic acid does not

20. Barium is below calcium in Group II of the periodic table. When these metals react with water, the main differences in observation would be the rate of reaction and

(A) solubility of product
(B) type of gas evolved
(C) reaction with litmus
(D) number of products formed

21. In which formula does hydrogen have a negative oxidation number?

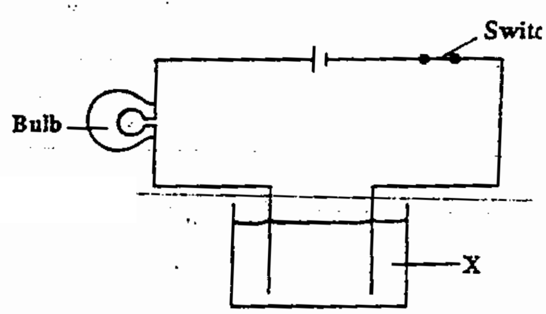
(A) CH_4

(B) H_2O_2

(C) NH_3

(D) NaH

<u>Item 22</u> refers to the following diagram.



22. The bulb will light the MOST brightly when X is

(A) a sugar solution
(B) distilled water
(C) aqueous ethanoic acid
(D) dilute sulphuric acid

23. Two solutions are mixed in order to demonstrate endothermic change. Which of the following techniques will be MOST appropriate?

(A) Taking mass readings
(B) Taking temperature readings
(C) Carefully observing colour changes
(D) Monitoring the pH of the solutions

24. The reaction occurring at the cathode during the electrolysis of copper sulphate using inert electrodes is given by the equation

(A) $Cu \rightarrow Cu^{2+} + 2e^-$
(B) $Cu^{2+} + 2e^- \rightarrow Cu$
(C) $2H^+ + 2e^- \rightarrow H_2$
(D) $4OH^- - 4e^- \rightarrow 2H_2O + O_2$

25. $^{24}_{11}Na$ and $^{23}_{11}Na$ are isotopes because they

(A) react vigorously with water
(B) have the same relative atomic mass
(C) have the same number of protons
(D) conduct electricity

26. When solid lead nitrate is heated, it decomposes giving off nitrogen (IV) oxide and oxygen. The balanced equation for this reaction is

(A) $2Pb(NO_3)_2(s) \rightarrow 2PbO(s) + 4NO_2(g) + O_2(g)$

(B) $Pb(NO_3)_2(s) \rightarrow PbO(s) + NO_2(g) + O_2(g)$

(C) $Pb(NO_3)_2(s) \rightarrow PbO(s) + 2NO_2(g) + O_2(g)$

(D) $2Pb(NO_3)_2(s) \rightarrow PbO(s) + 2NO_2(g) + O_2(g)$

27. Which of the following halogens is a liquid at room temperature?

(A) Bromine
(B) Chlorine
(C) Fluorine
(D) Iodine

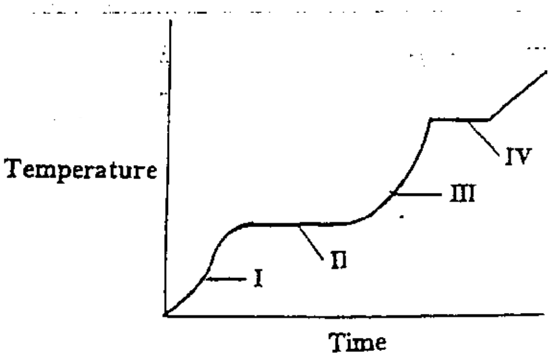
28. A solution has a pH of 1. This solution would be expected to

(A) react with zinc metal to produce hydrogen
(B) react with zinc metal to produce a solution of pH 10
(C) neutralise a solution of pH 4
(D) react with hydrochloric acid to produce a salt and water

29. 'Ionic bond formation' is BEST described as the result of the

(A) sharing of electrons between the atoms of a metal and non-metal to achieve stability
(B) donation of electrons from a metal to a non-metal to achieve stability
(C) donation of electrons from a non-metal to a metal to achieve stability
(D) attraction between the positively charged ions of a metal and a pool of electrons

Item 30 refers to the following graph which shows the changes in temperature with time, for a substance heated until no further physical changes take place.



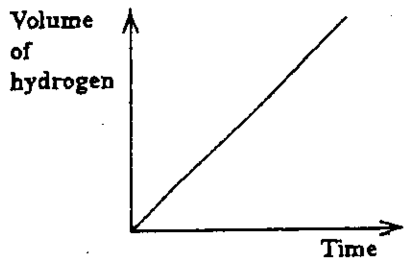
30. During which sections of the graph is the substance changing its state?

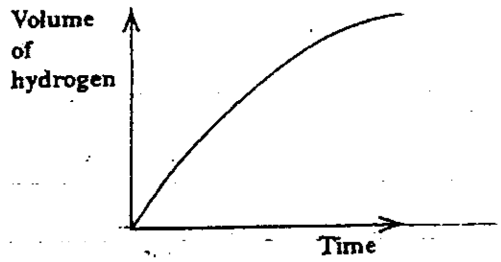
(A) I and II
(B) I and III
(C) II and IV
(D) III and IV

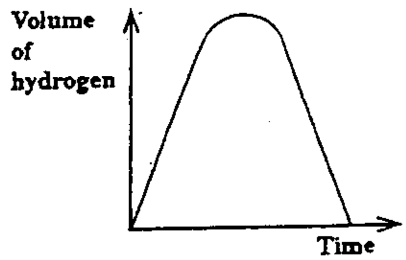
31. When heat is given off to the surroundings during a chemical reaction, it is because bond breaking

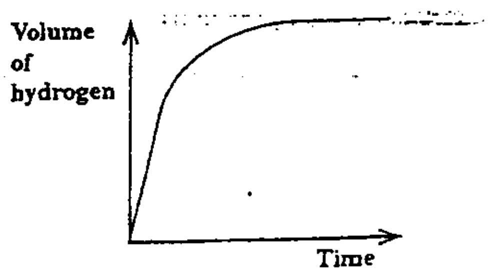
(A) releases energy as well as bond making
(B) requires less energy than is released when new bonds are formed
(C) requires more energy than that released when new bonds are formed
(D) is an endothermic process whereas bond making is an exothermic one

32. An excess of magnesium powder is added to 50 cm^3 of dilute sulphuric acid, and the reaction is allowed to continue until no more hydrogen evolves. Which of the following graphs BEST represents the complete reaction?

(A)


(B)


(C)


(D)


33. A lump of metal is reacted with an acid to produce hydrogen gas. Which of the following procedures could be employed in order to increase the rate of the reaction?

I. Increasing the temperature at which the reaction is carried out
II. Subdividing the lump of metal
III. Reducing the concentration of the acid

(A) I and III only
(B) I and II only
(C) II and III only
(D) I, II and III

34. When 100 cm^3 of 2 mol dm^{-3} hydrochloric acid (an excess) are added to 5.0g of granulated zinc, hydrogen is evolved at a certain rate. If 5.0g of powdered zinc were used instead, the rate of production of hydrogen would ——————————

(A) increase because of the greater concentration of zinc
(B) increase because of the increased surface area of zinc
(C) decrease because of the decreased reactivity of zinc powder
(D) decrease because of the lower solubility of powdered zinc

35. Which of the following aqueous solutions contains 1 mole of hydrogen ions?

(A) 1 dm^3 of 1.0 mol dm^{-3} H_2SO_4
(B) 1 dm^3 of 1 mol dm^{-3} CH_3COOH
(C) 2 dm^3 of 0.5 mol dm^{-3} H_2SO_4
(D) 2 dm^3 of 0.5 mol dm^{-3} HCl

36. Ethanol can react with concentrated sulphuric acid to produce ethene. This reaction is referred to as

(A) dehydration
(B) esterification
(C) dehydrogenation
(D) neutralisation

37. Which of the following equations represents a condensation reaction?

(A) $nC_6H_{12}O_6 \rightarrow (C_6H_{10}O_5)n + nH_2O$

(B) $CH_3CH = CH_2 + HBr \rightarrow CH_3 - CH_2 - CH_2Br$

(C) $CH_3CH_2CH_3 + Br_2 \rightarrow CH_3CH_2CH_2Br + HBr$

(D) $CH_3CH_2COOH + NaOH \rightarrow CH_3CH_2COONa + H_2O$

38. Which of the following compounds is NOT a member of the alkene series?

(A) C_2H_4

(B) C_5H_{10}

(C) C_2H_6

(D) C_4H_8

39. Octane, the major component of gasoline, is an alkane with 8 carbon atoms per molecule. Which of the following is the formula of octane?

(A) C_8H_{12}
(B) C_8H_{14}
(C) C_8H_{16}
(D) C_8H_{18}

40. The compound ethene is described as being unsaturated. This means that the

(A) carbon atoms in ethene are linked by single bonds
(B) molecule has insufficient hydrogen atoms
(C) carbon atoms in the molecule are very reactive
(D) molecule contains at least one double bond

Item 41 refers to the following equation.

$$\text{Ethene} \xrightarrow[\text{conc. } H_2SO_4]{H_2O} \text{I}$$

41. The product I is

(A) ethanol
(B) ethane
(C) ethanoic acid
(D) ether

42. The number of isomers of butane C_4H_{10} is

(A) 1
(B) 2
(C) 3
(D) 4

Items 43 - 44 refer to the following substances

(A) Natural gas
(B) Cyclohexene
(C) Ethanol
(D) Vinegar

Match each item below with one of the substances above. Each substance may be used once, more than once or not at all to answer the items below.

43. Decolourises bromine water

44. Is an unsaturated hydrocarbon

45. Ethanol may be used as a fuel because it

(A) has a high heat of combustion
(B) is easily converted to carbon
(C) is easily converted to ethane
(D) has a low heat of formation

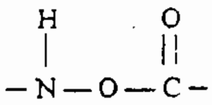
46. Glucose is converted to starch or cellulose by

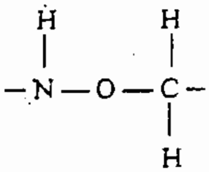
(A) addition polymerisation
(B) oxidation and reduction
(C) condensation polymerisation
(D) dehydrogenation

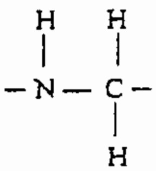
47. The following represents a portion of the nylon molecule (a polyamide).

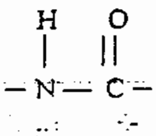
-X-Y-X-Y-

X and Y represent the monomers minus the functional groups, and "-" represents the linkage. The linkage could BEST be represented by

(A) 

(B) 

(C) 

(D) 

48. Which of the following substances give a blue colouration with iodine?

(A) Fats
(B) Starches
(C) Proteins
(D) Polyamides

49. Hydrocarbons are obtained naturally from

(A) petroleum
(B) methane
(C) carbon
(D) hydrogen

50. During the manufacture of ethanol by fermentation, the gas evolved

(A) relights a glowing splint
(B) burns with a blue flame
(C) turns acidified potassium dichromate green
(D) turns lime water cloudy

51. The reaction of fats with concentrated aqueous sodium hydroxide is commonly referred to as

(A) esterification
(B) saponification
(C) neutralisation
(D) condensation

52. During the digestion of polysaccharide by the human body, the main process occurring is

(A) hydrolysis
(B) condensation
(C) neutralisation
(D) polymerisation

53. Which of the following may be true of metals?

I. They form solid chlorides.
II. They generally form basic oxides.
III. They conduct electricity only when molten.

(A) III only
(B) I and II only
(C) II and III only
(D) I, II and III

54. What is observed when EXCESS aqueous ammonia is added to a solution of copper (II) sulphate?

(A) Light blue precipitate
(B) Deep blue solution
(C) Green precipitate
(D) Green solution

Items 55 and 56 refer to the following equation.

$$N_2(g) + 3H_2(g) = 2NH_3(g), \Delta H = -92 \text{ kJ mol}^{-1}$$

55. The role of the catalyst in this reaction is to

(A) increase the yield of ammonia in the mixture
(B) lower the energy level of the reactants
(C) increase the rate of production of ammonia
(D) decrease the amount of heat lost

56. The notation $\Delta H = -92 \text{kJmol}^{-1}$ means that

(A) 92 kJ of energy are required for the reaction to proceed
(B) 92 kJ of energy are released during the reaction
(C) 92 kJ of energy are required to break the bonds in N_2 and H_2
(D) 92 kJ of energy are required to form the bonds in NH_3

57. Which of the following substances may be responsible for acid rain?

(A) Dust particles
(B) Radioactive fallout
(C) Carbon dioxide and carbon monoxide
(D) Sulphur dioxide and hydrogen sulphide

58. Which of the following processes gives a different gaseous product from the others?

(A) Photosynthesis in plants
(B) Respiration in animals
(C) Oxidation of carbon monoxide
(D) Heating calcium carbonate

59. Aluminium is obtained industrially by

(A) reducing its oxide in the blast furnace
(B) using magnesium to displace it from aqueous solution
(C) the electrolysis of molten cryolite
(D) the electrolysis of molten alumina

60. Both glass and ceramics contain the non-metal

(A) phosphorus
(B) nitrogen
(C) silicon
(D) sulphur

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.